Programming and Algorithms

COMP1038.PGA

Week 9 – Lecture 3: Trees and Graphs

Dr. Pushpendu Kar

Overview

- Tree
- Graph
  - Dijkstra's Algorithm

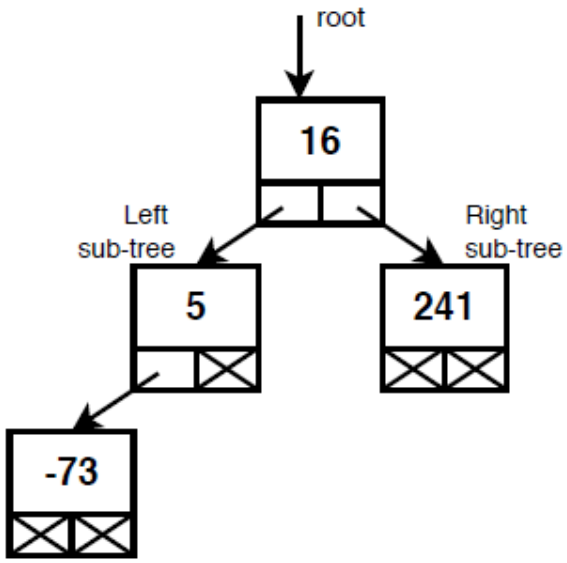
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Tree

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Tree: introduction

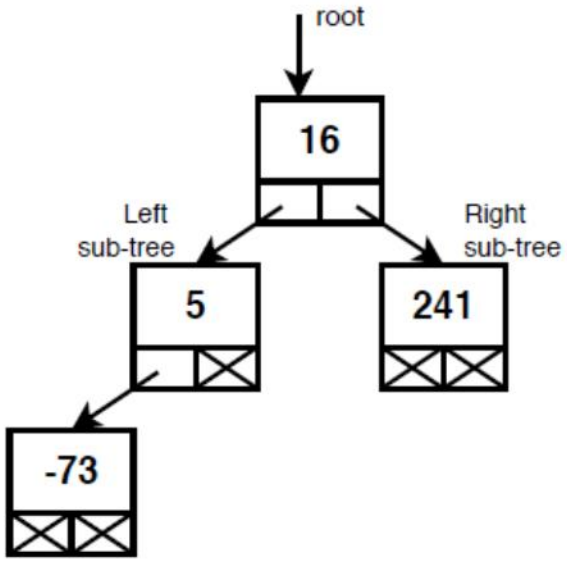
- Trees are hierarchical data structures containing nodes which store a value and references to 2 or more subtrees.
- The "start" of the tree is the root node.
- Nodes with no subtrees are called leaf nodes.
- A binary tree is a tree where each node has exactly two possible children.



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Tree: binary search tree

- BSTs are binary trees in which the values are stored in the tree in some specified order.
- Eg, for an integer BST,
  - every value in the left sub-tree < value in the node
  - every value in the right sub-tree >= the value in the node.
- Searching for values in a BST can be extremely quick because each comparison discards half the remaining values (on average).
- Inserting/removing nodes is more complex as it may require moving existing nodes.



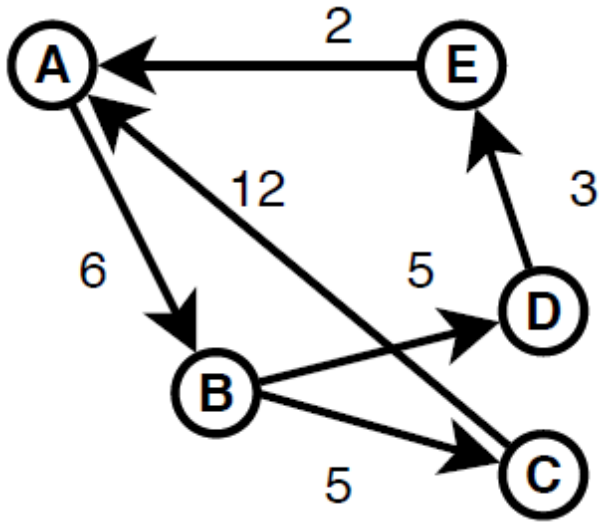
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Graph

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Graph: introduction

- A graph represents a set of vertices (or nodes) and a set of edges which are connections between vertices.
- Edges can be directed or undirected.
- Edges can be weighted or unweighted.
- Graphs can be connected (for any 2 vertices, there is a path between them) or unconnected.
- …and many other possible properties
  - Graph theory is a major branch of discrete maths.



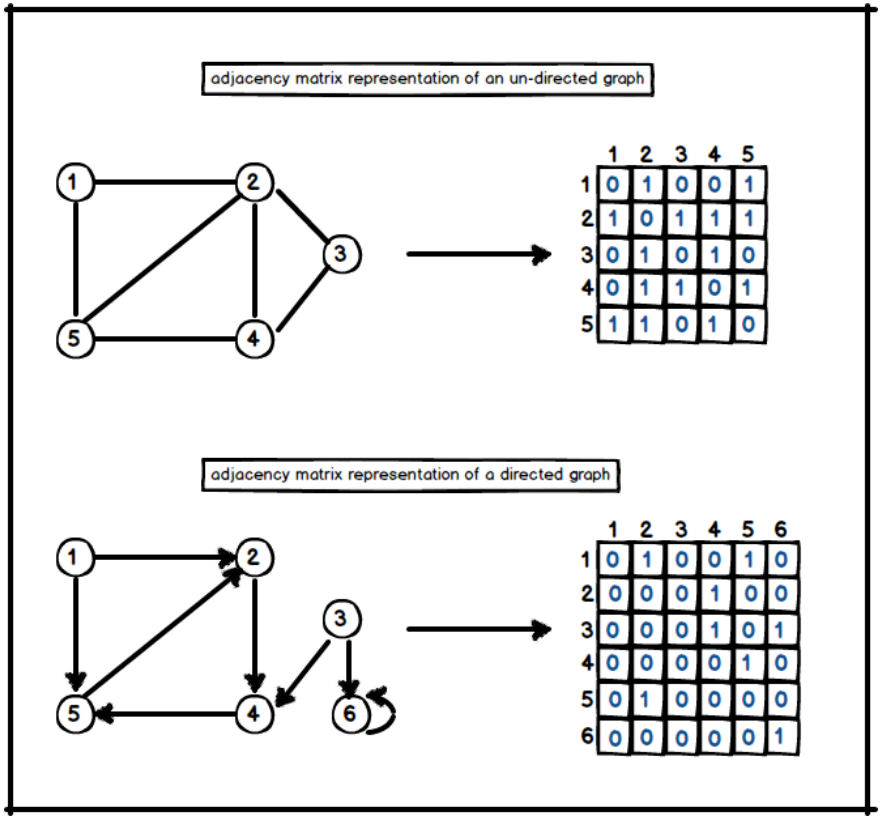
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Graph: adjacency matrix

- Vertices are assigned to integer IDs.
- Store graph as a 2D array of integers.
- A_{ij} contains a value if there is an edge from i to j.
  - Store edge present? Number of edges? Weight of edge?
- Very fast look-up for edge between two vertices.
- Low memory usage but wastes lots of space for sparse graphs.
- Requires whole array to be changed when adding/removing vertices.

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Graph: adjacency matrix cont…



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Graph: applications

- Modeling relationships between users in social media networks (vertex=user, edge=friends).
- Navigation through cities vertex=road junction, edge=road between junction, weight=distance).
- Modelling state machines (vertex=state, edge=valid transition).

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Introduction to Dijkstra's Algorithm

- **Dijkstra's algorithm** - is a solution to the single-source shortest path problem in graph theory.
- Works on both directed and undirected graphs. However, all edges must have nonnegative weights.
- Approach: Greedy
- Input: Weighted graph G={E,V} and source vertex $v \in V$, such that all edge weights are nonnegative
- Output: Lengths of shortest paths (or the shortest paths themselves) from a given source vertex $v \in V$ to all other vertices

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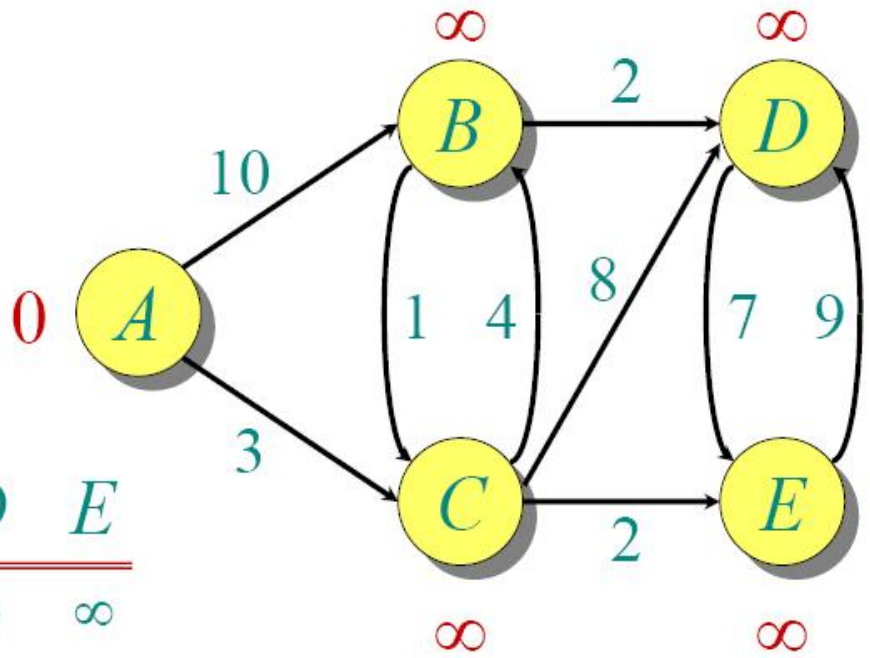
Dijkstra's Algorithm: Pseudocode

```
dist[s] ←0                                  (distance to source vertex is zero)
for  all v ∈ V–{s}
       do  dist[v] ←∞                       (set all other distances to infinity)
S←∅                                         (S, the set of visited vertices is initially empty)
Q←V                                         (Q, the queue initially contains all vertices)
while Q ≠∅                                  (while the queue is not empty)
do   u ← mindistance(Q,dist)                (select the element of Q with the min. distance)
     S←S∪{u}                                (add u to list of visited vertices)
      for all v ∈ neighbors[u]
            do  if   dist[v] > dist[u] + w(u, v)        (if new shortest path found)
                then      d[v] ←d[u] + w(u, v)          (set new value of shortest path)
                                                        (if desired, add traceback code)
return dist
```


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Dijkstra Example

Initialize:

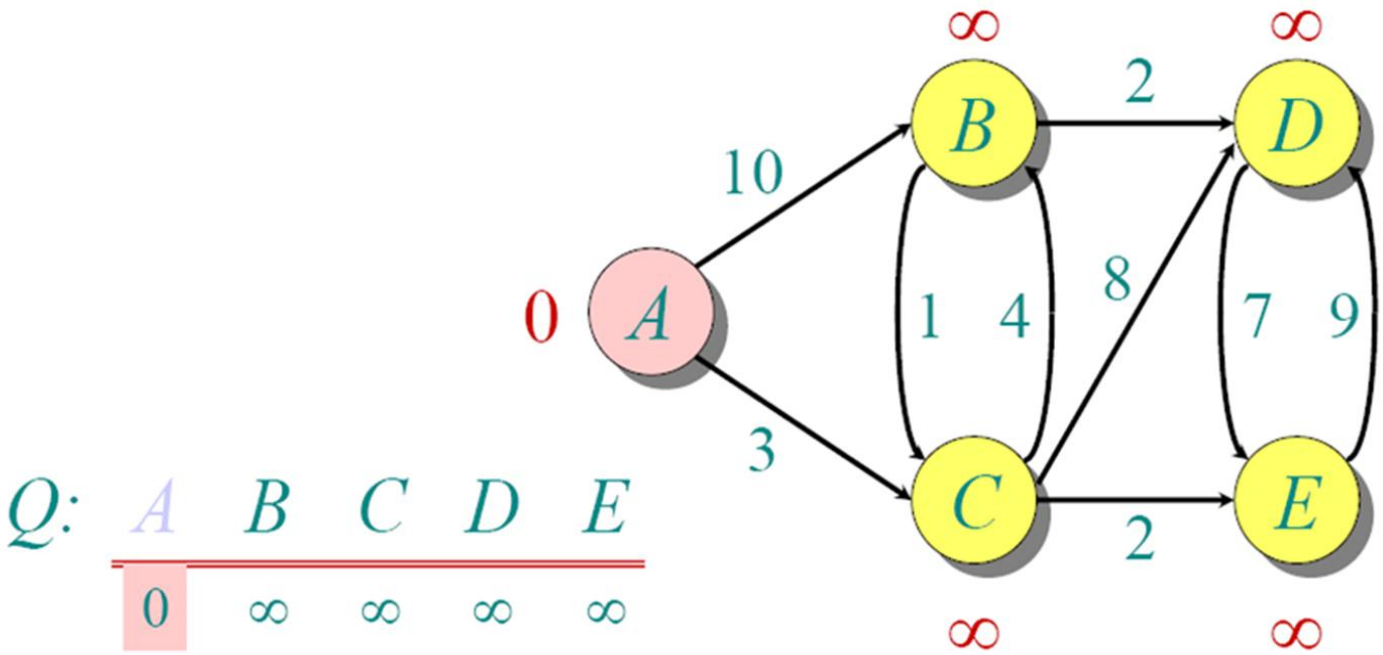


Q:	A	B	C	D	E
	0	∞	∞	∞	∞

S: {}

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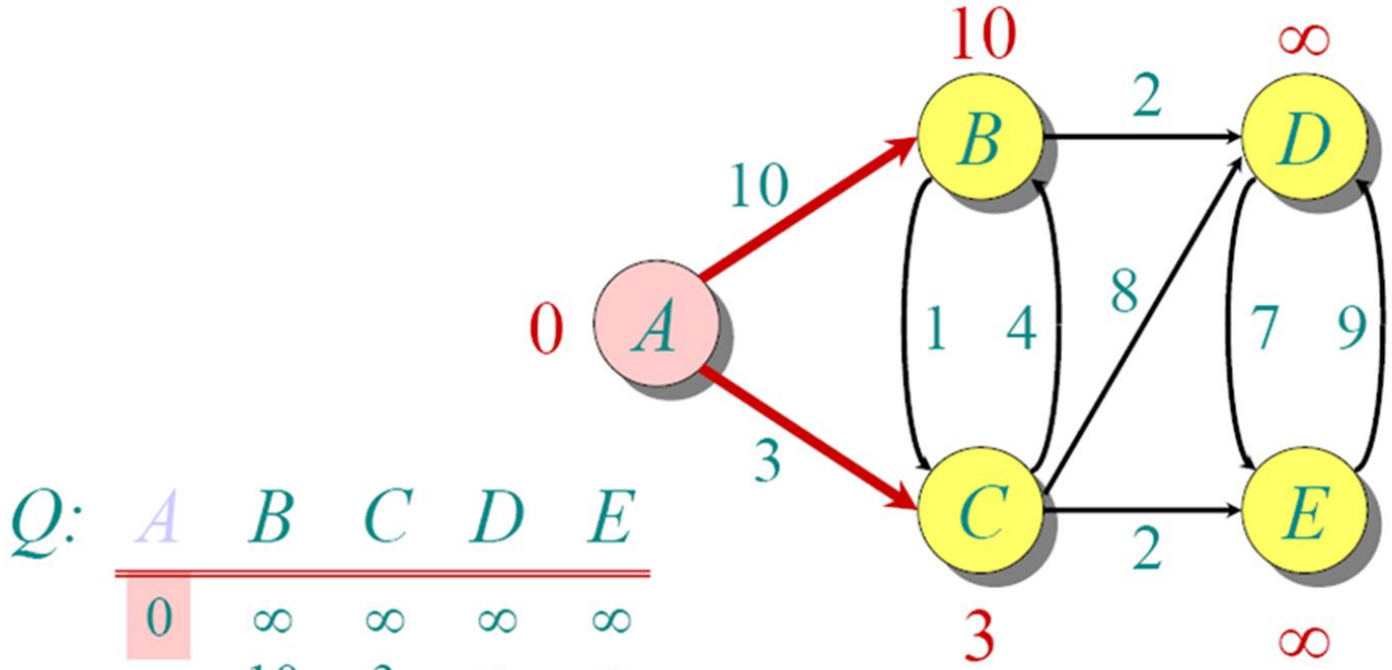
Dijkstra Example



Q:	A	B	C	D	E
	0	∞	∞	∞	∞

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Dijkstra Example

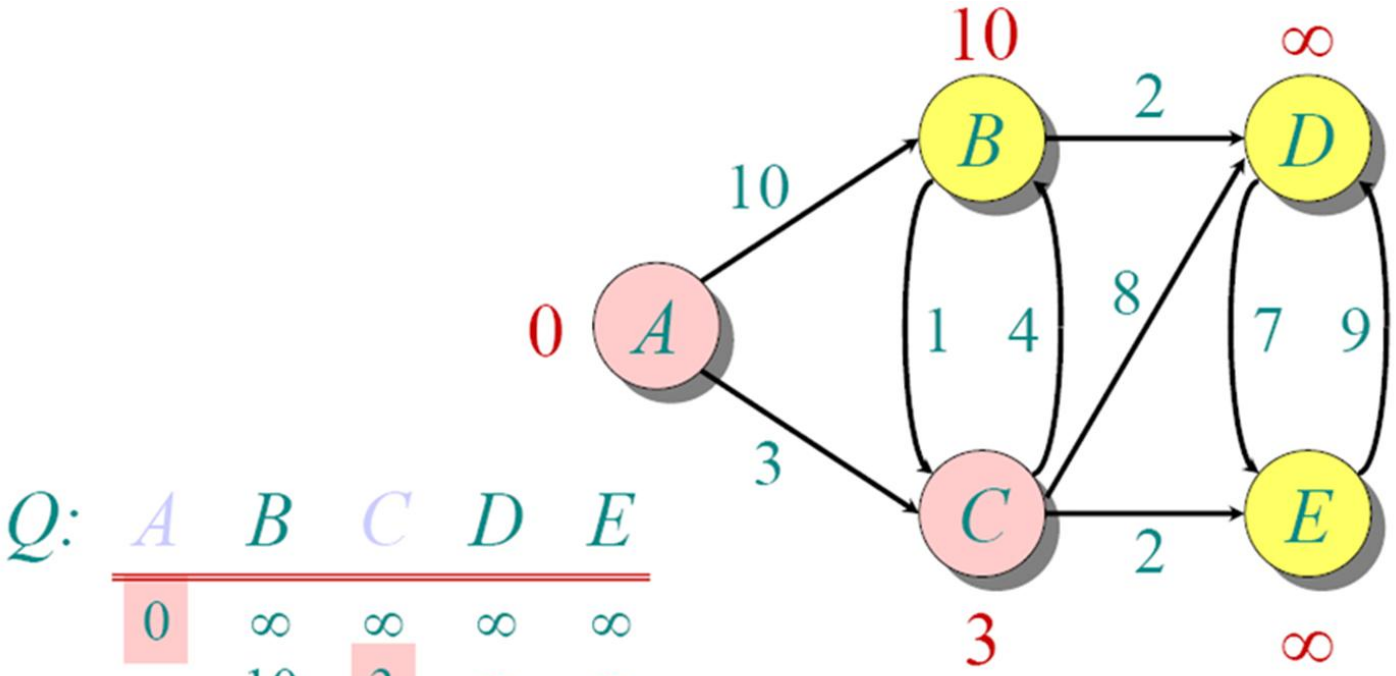


Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞

$S: \{ A \}$

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Dijkstra Example

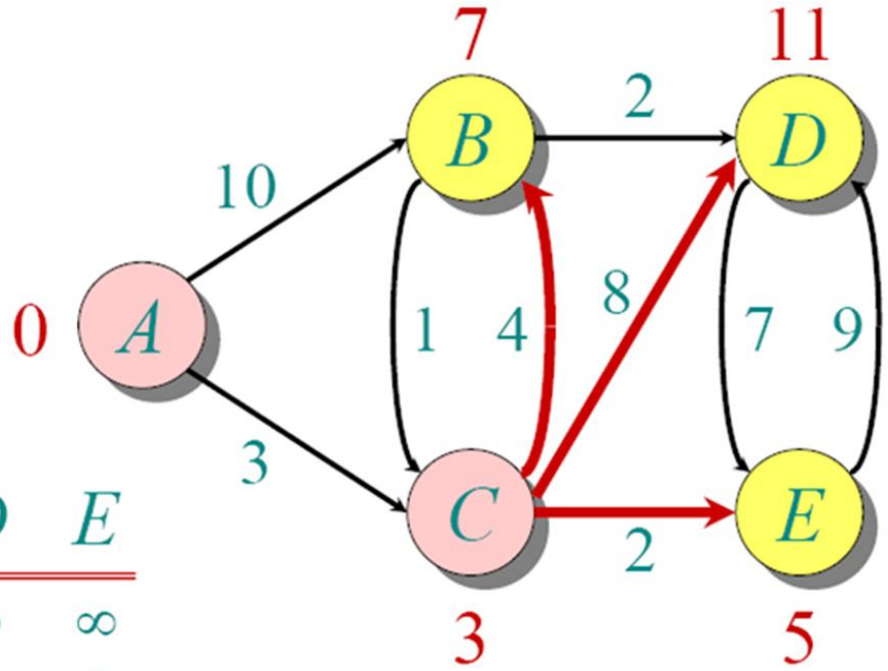


Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞

$S: \{ A, C \}$

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Dijkstra Example

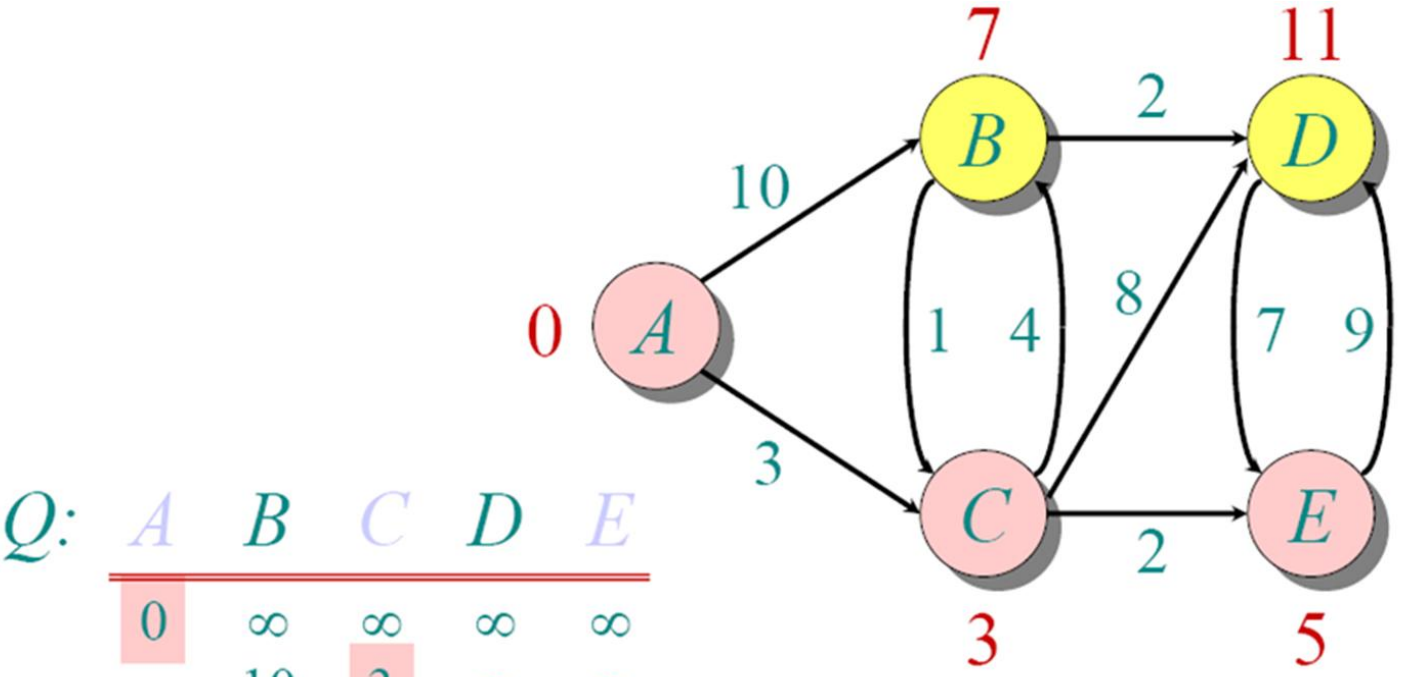


Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5

$S: \{ A, C \}$

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Dijkstra Example

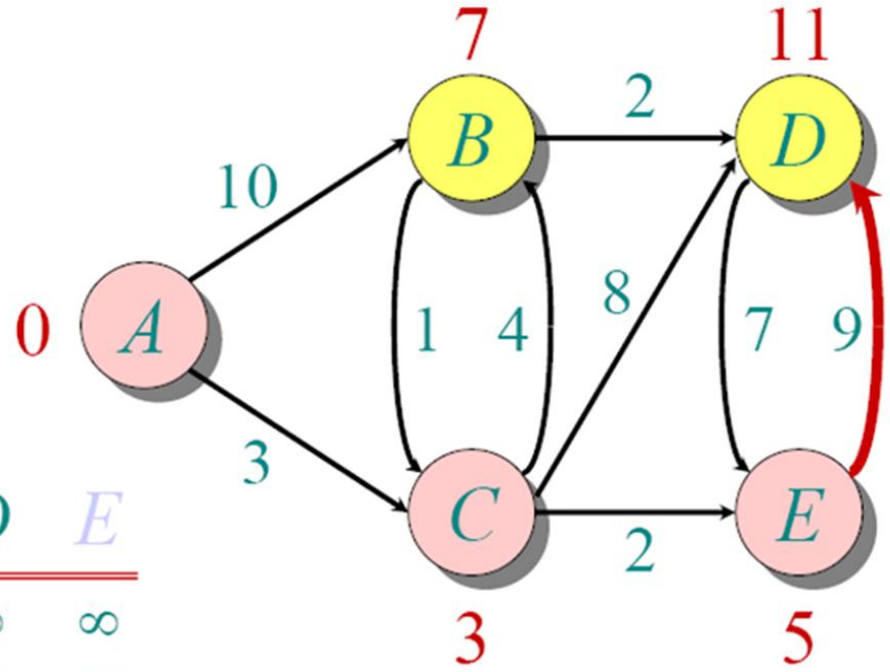


Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5

$S: \{ A, C, E \}$

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Dijkstra Example

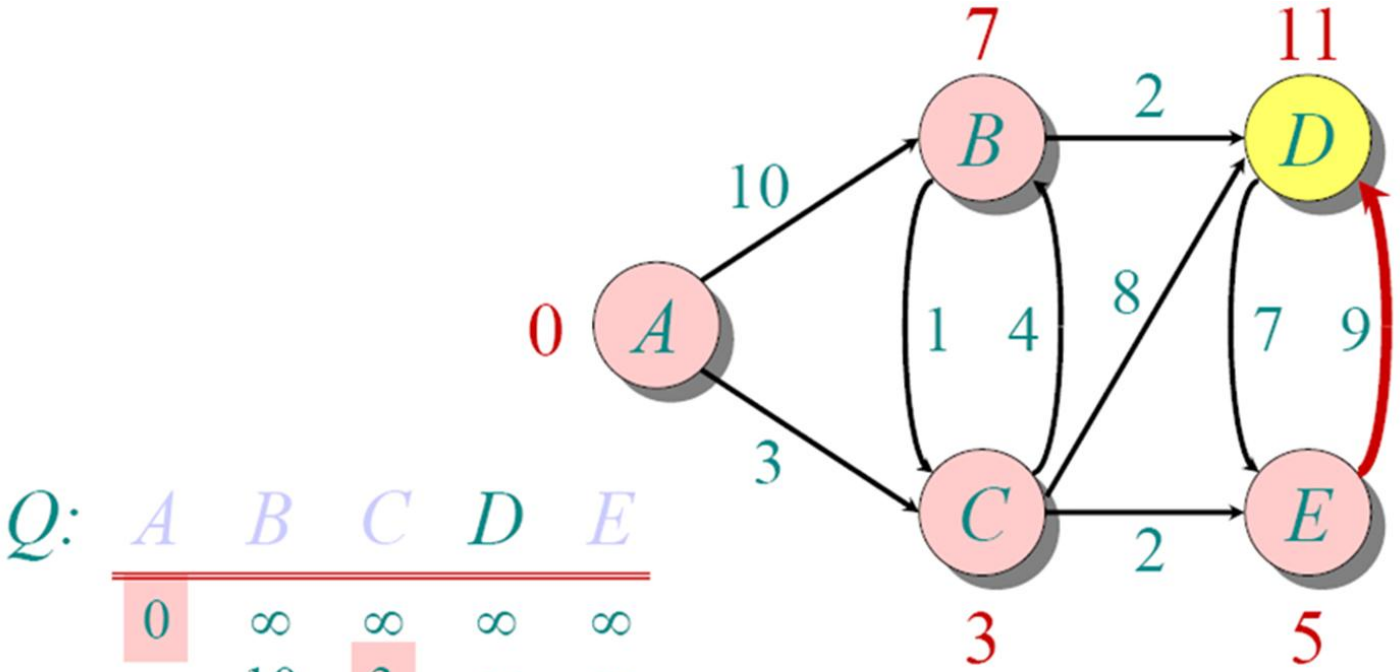


Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	

$S: \{ A, C, E \}$

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Dijkstra Example

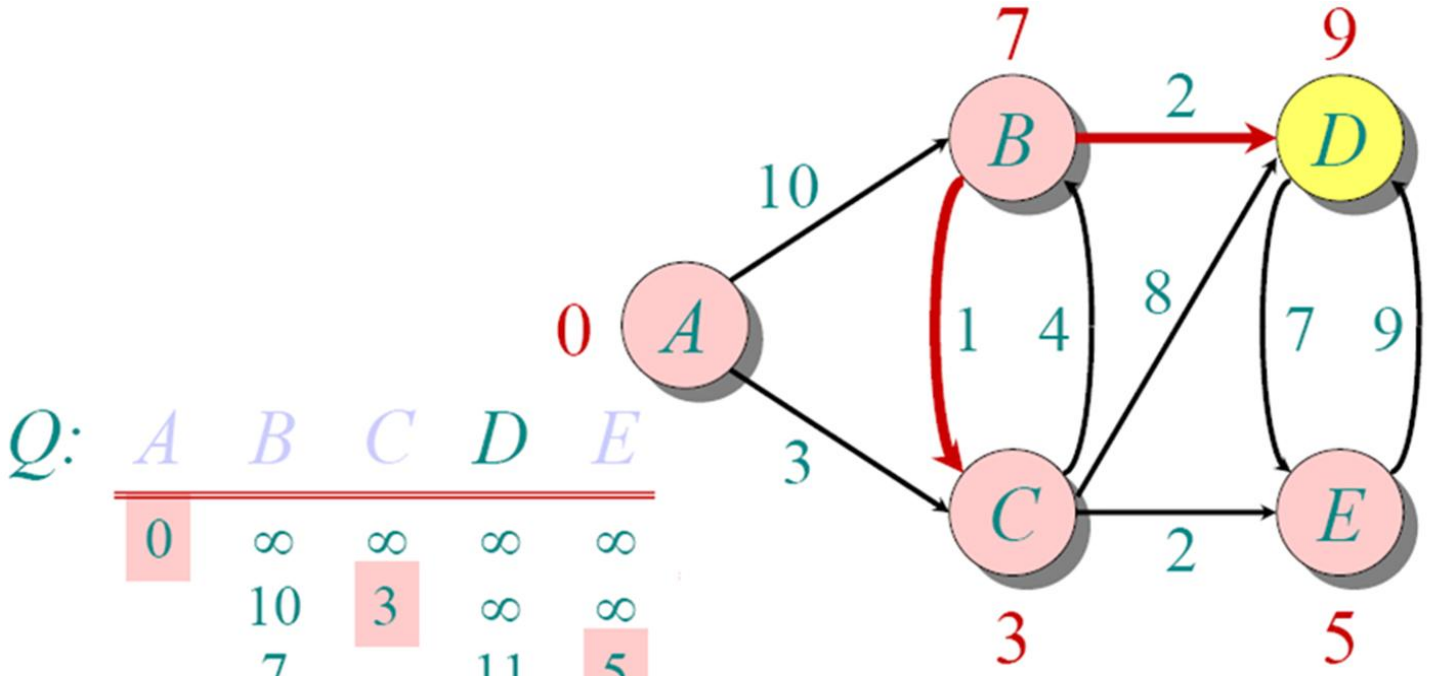


Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	

$S: \{ A, C, E, B \}$

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Dijkstra Example

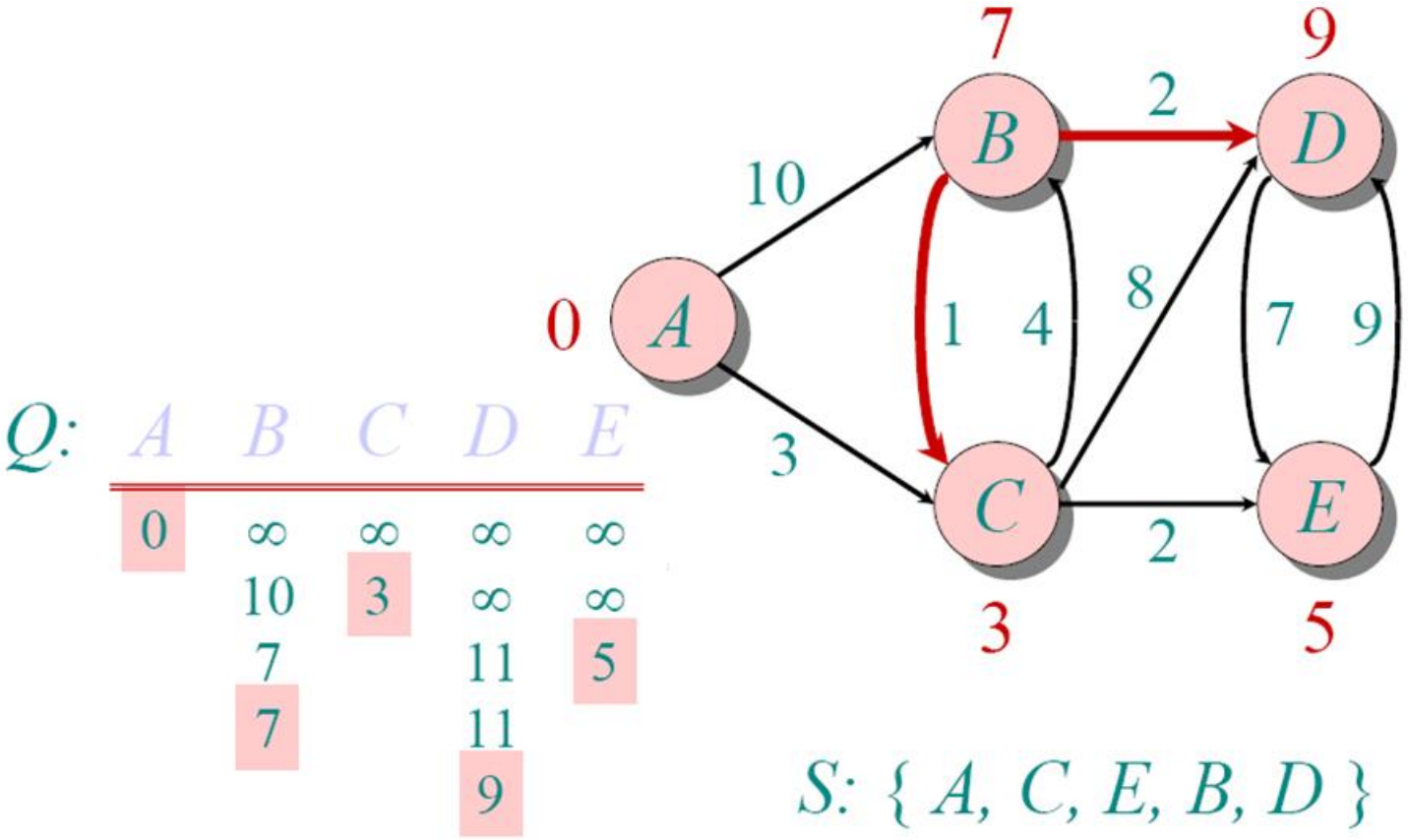


Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	
				9	

$S: \{ A, C, E, B \}$

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Dijkstra Example



Q:

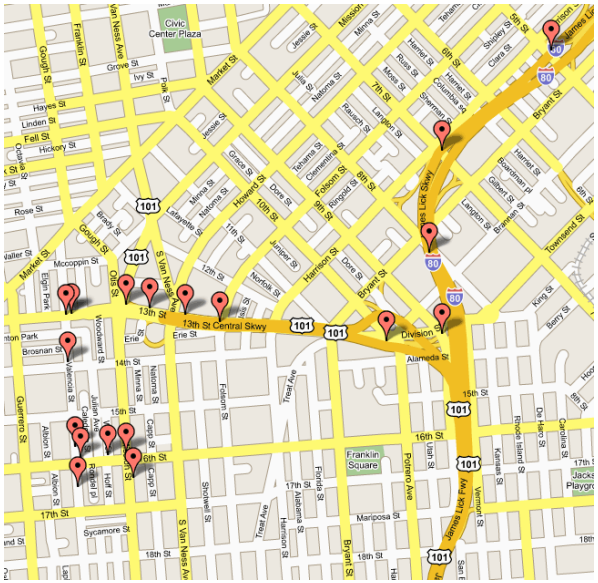
A	B	C	D	E
0	∞	∞	∞	∞
	10	3	∞	∞
	7		11	5
	7		11	
			9	

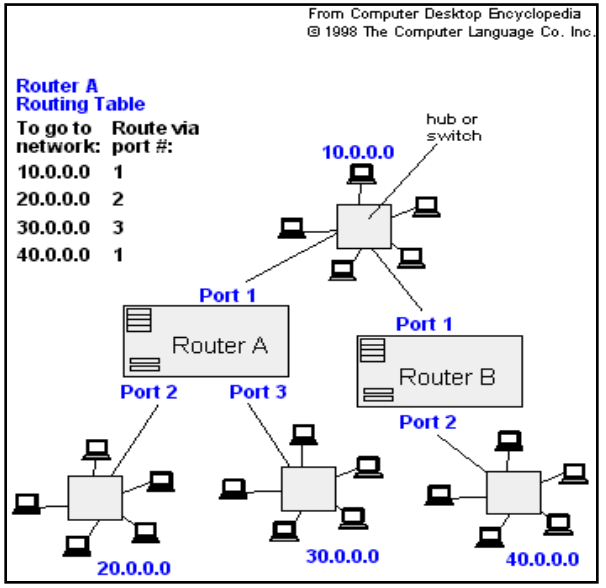
S: { A, C, E, B, D }

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Applications of Dijkstra's Algorithm

- Traffic Information Systems are most prominent use.
- Mapping (Map Quest, Google Maps)
- Routing Systems





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Thank you!

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